1.

Suppose a horizontal block-spring system has a spring constant 1,178 N/m and block of mass 2.7 kg. Calculate the frequency in Hz of its harmonic motion.

2.

Suppose a horizontal block-spring system has a spring constant 2,199 N/m and block of mass 7.3 kg. Calculate the period in s of its harmonic motion.

3.

A tunnel is drilled straight through the center of an airless planet to the other side. For simplicity, the planet is assumed to have constant density. The equation for the motion of an object falling through the tunnel is given by

$$\frac{d^2 r}{dt^2} + \left(\frac{4\pi G\rho}{3}\right) r = 0$$

where G = 6.67 × 10^{-11} $m^3/(kg{\cdot}s^2)$ is the gravitation constant. The planet's density and radius are ρ = 3,743 kg/m^3 and R = 208 km.

If an astronaut jumps in the tunnel, how many ***hours*** would it take for her to fall all the way to the other side and return to her original position?

T = _____ h (Note the units!)

4.

Suppose a harmonic oscillator is described by $ma_z + kz = 0$ where m = 18.5 kg and k = 66 N/m. Calculate the period of the harmonic motion.

5.

Fill In the Blanks with numerical values. ***Give three digits for each answer***, rounding as appropriate. For example, if your answer is "3", report "3.00". If your answer is "1.5555", report "1.56". If your answer is "0.7", report "0.700".

Suppose a mass-spring system has the description $x(t) = 5.50 \cos(3.00t - 1.40)$, with all units in the MKS system.

The amplitude is _____ m, the angular frequency is _____ rad/s, the phase shift is _____ rad, the frequency is _____ Hz; the period is _____ s.

6.

Choose the correct answers.

Suppose a simple harmonic oscillator solution for a mass-spring system is given by $x(t) = 0.7 \cos(8t + 1.9)$, where all MKS units are assumed.

For this solution,

(a) the amplitude is [Select] m;

(b) the angular frequency is [Select] rad/s;

(c) the phase shift is [Select] rad;

(d) the frequency is [Select] Hz;

(e) the period is [Select] s;

(f) the maximum speed is [Select] m/s;

(g) the maximum acceleration is [Select] m/s^2;

(h) the position at t = 0 s is [Select] m;

(i) the velocity at t = 0 s is [Select] m/s;

(j) the acceleration at t = 0 s is [Select] m/s^2;

(k) Find the first time t > 0 that the mass reaches x = 0.

[Select]

(l) Find the first time t > 0 that the speed is 0.

[Select]

7.

Suppose a mass-spring system has spring constant 456 N/m and mass 5 kg. The mass is displaced in the positive x-direction by 0.5 m and set in motion at 2.4 m/s in the negative x-direction. Calculate the phase shift as a positive angle in radians, assuming a cosine solution.

8.

A simple harmonic oscillator consists of a block of unknown mass and spring with spring constant 114 N/m on a frictionless horizontal surface. The block is moved 0.27 m from equilibrium, stretching the spring, and released from rest. If the block's maximum acceleration is 3.4 m/s^2, calculate the mass of the block in kg.

9.

A simple harmonic oscillator consists of a block of mass 8 kg and a horizontal spring with constant 186 N/m. If the block is moved 0.34 m from equilibrium, stretching the spring, calculate the block's maximum acceleration a_{max}.

a_{max} = _____ m/s^2

10.

A 7-kg block attached to a spring of force constant 132 N/m undergoes simple harmonic motion. At a particular time the mass' speed is 1.92 m/s and its position is 0.27 m. Calculate the amplitude of its oscillation.

A = _____m

11.

A 0.48-kg coffee cup is attached to a suspended spring and slowly lowered, stretching the spring by 3 cm. From this equilibrium position, the cup is pulled down 6 cm and released from rest. Find the cup's speed as it passes through its equilibrium position.

v = _____m/s

12.

A simple pendulum has length 0.97 m and mass 6.6 kg. If the local acceleration of gravity on this planet is 11.4 m/s^2, calculate the period of the motion.

T = _____ s

13.

A simple pendulum has length 1.8 m, mass 8.1 kg. If the local acceleration of gravity on this planet is 7.89 m/s^2, calculate the frequency of the motion in Hz.

14.

Suppose an irregular chunk of matter of mass 18.3 kg has a moment of inertia 9.45 $kg{\cdot}m^2$ about its pivot located 1.75 m away from its center of mass. Calculate the angular frequency in rad/s of small oscillations.

15.

An object is free to oscillate around a pivot located a distance 1.34 m from its center of mass. Calculate the period of small oscillations if its mass is 7.2 kg and its moment of inertia about the pivot is 15.7 kg·m^2.

T = _____ s

16.

S

A simple harmonic oscillator oscillates with amplitude A around its equilibrium point at x = 0. The maximum speed occurs at

[Select] , the maximum magnitude acceleration

occurs at x = A or x = -A , and the total energy is

[Select] .